Folded-Cascode Op-Amp Design

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1 Introduction and Circuit Configuration

The NMOS-input folded-cascode op-amp is sized with the nine-step design approach of **P. E. Allen, Lecture 24** (Example 24-4). The transistor names of the schematic map to Allen's names as follows:

Schematic	Allen (Lecture 24)	Function
M_0, M_1	M_1, M_2	NMOS input differential pair
M_2	M_3	NMOS tail current source
M_3, M_4	M_4, M_5	Top PMOS current sources
M_5, M_6	M_6, M_7	PMOS cascodes
M_7, M_8	M_8, M_9	NMOS cascodes
M_9, M_{10}	M_{10}, M_{11}	Bottom NMOS current mirror

Note: the aspect ratio is $S_i = (W/L)_i$. The schematic names ($M_0 \ldots M_{10}$) are used throughout this document.

1.1 Technology Parameters

- $V_{DD} = 1.2$ V, $V_{SS} = 0$ V, $L = 0.5$ μm.
- $K'_N = 457.2$ μA/V^2, $K'_P = 63.8$ μA/V^2.
- $V_{TN0} = 0.343$ V, $|V_{TP0}| = 0.235$ V.
- $\lambda_N = 0.0528$ V^{-1}, $\lambda_P = 0.0469$ V^{-1}.

1.2 Target Specifications

1. Slew rate, $SR \geq 50$ V/μs.
2. Load capacitance, $C_L = 2$ pF.
3. Gain-bandwidth, $GB = 50$ MHz.
4. Output voltage swing: $V_{out(min)} = 0.3$ V, $V_{out(max)} = 0.9$ V.
5. Input common-mode range (ICMR): $V_{IC(min)} = 0.85$ V, $V_{IC(max)} = 1.05$ V.
6. DC voltage gain, $A_v(0) \geq 5000$ V/V (≈ 74 dB).

2 Nine-Step Design Procedure (Example 24-4)

2.1 Step 1: Slew Rate

The tail current I_{D2} (Allen's I_3) is set by

$$I_{D2} = SR \cdot C_L = (50 \times 10^6) \times (2 \times 10^{-12}) = 100\ \mu\text{A} \tag{1}$$

$$\boxed{I_{D2} = 100\ \mu\text{A}, \quad I_{D0} = I_{D1} = \frac{I_{D2}}{2} = 50\ \mu\text{A}} \tag{2}$$

2.2 Step 2: Bias Currents in the Output Cascodes

The branch currents are chosen as $I_{D3} = I_{D4} = 1.25 \times I_{D2}$ to avoid zero current in the cascodes. *The factor 1.25 is the same as in Example 24-4.*

$$\boxed{I_{D3} = I_{D4} = 1.25 \times 100\mu = 125\ \mu\text{A}} \tag{3}$$

The quiescent current in each cascode branch is the difference between the branch current and the input-pair current:

$$I_{casc} = I_{D3} - I_{D0} = 125\mu - 50\mu = 75\ \mu\text{A} \tag{4}$$

2.3 Step 3: Maximum Output Voltage, v_{out}(max)

The upper headroom is split equally between M_3 (or M_4) and M_5 (or M_6):

$$V_{SD3(sat)} = V_{SD5(sat)} = 0.5 \times (V_{DD} - V_{out(max)}) = 0.5 \times (1.2 - 0.9) = 0.15\ \text{V} \tag{5}$$

Sizing of M_3, M_4 (top current sources):

$$S_3 = S_4 = \frac{2I_{D3}}{K'_P (V_{SD3(sat)})^2} = \frac{2 \times 125\mu}{63.8\mu \times (0.15)^2} = \frac{250}{1.4355} \approx 174.14 \tag{6}$$

For M_5 and M_6 we assume the **worst-case** current, i.e. the whole I_{D3} flows through the cascode when the input pair is fully unbalanced, so the same $I_{D3} = 125\ \mu\text{A}$ is used:

$$S_5 = S_6 = \frac{2 \times 125\mu}{63.8\mu \times (0.15)^2} = 174.14 \tag{7}$$

$$\boxed{S_3 = S_4 = S_5 = S_6 = 174.14 \implies W_3 = W_4 = W_5 = W_6 = 174.14 \times 0.5 \approx 87.1\ \mu\text{m}} \tag{8}$$

2.4 Step 4: Minimum Output Voltage, v_{out}(min)

The lower headroom is split equally between M_9 (or M_{10}) and M_7 (or M_8):

$$V_{DS9(sat)} = V_{DS7(sat)} = 0.5 \times (V_{out(min)} - V_{SS}) = 0.5 \times (0.3 - 0) = 0.15\ \text{V} \tag{9}$$

As in Step 3, the **worst-case** current $I_{D3} = 125\ \mu\text{A}$ is used for all four NMOS transistors:

$$S_7 = S_8 = S_9 = S_{10} = \frac{2 \times 125\mu}{457.2\mu \times (0.15)^2} = \frac{250}{10.287} \approx 24.30 \tag{10}$$

$$\boxed{S_7 = S_8 = S_9 = S_{10} = 24.30 \implies W_7 = W_8 = W_9 = W_{10} = 24.30 \times 0.5 \approx 12.15\ \mu\text{m}} \tag{11}$$

2.5 Step 5: Gain-Bandwidth ($GB = g_{m0}/C_L$)

$$S_0 = S_1 = \frac{GB^2 C_L^2}{K_N' I_{D2}} = \frac{(2\pi \times 50 \times 10^6)^2 \times (2 \times 10^{-12})^2}{457.2 \times 10^{-6} \times 100 \times 10^{-6}}$$
$$= \frac{(628.32 \times 10^{-6})^2}{45.72 \times 10^{-9}} = \frac{394.8 \times 10^{-9}}{45.72 \times 10^{-9}} \approx 8.63 \tag{12}$$

$$\boxed{S_0 = S_1 = 8.63 \implies W_0 = W_1 = 8.63 \times 0.5 = 4.32\ \mu\text{m}} \tag{13}$$

2.6 Step 6: Minimum Input Common-Mode Voltage ($V_{IC(min)}$)

From the nine-step table:

$$S_2 = \frac{2I_{D2}}{K_N' \left(V_{IC(min)} - V_{SS} - \sqrt{\dfrac{I_{D2}}{K_N' S_0}} - V_{TN0} \right)^2} \tag{14}$$

The square-root term is

$$\sqrt{\frac{I_{D2}}{K_N' S_0}} = \sqrt{\frac{100 \times 10^{-6}}{457.2 \times 10^{-6} \times 8.63}} = \sqrt{\frac{100}{3945.6}} = 0.159\ \text{V} \tag{15}$$

Substituting:

$$S_2 = \frac{200 \times 10^{-6}}{457.2 \times 10^{-6} \times (0.85 - 0 - 0.159 - 0.343)^2} = \frac{200}{457.2 \times (0.348)^2} = \frac{200}{55.38} \approx 3.61 \tag{16}$$

$$\boxed{S_2 = 3.61 \implies W_2 = 3.61 \times 0.5 = 1.81\ \mu\text{m}} \tag{17}$$

2.7 Step 7: Maximum Input Common-Mode Voltage ($V_{IC(max)}$)

Check whether S_3 from Step 3 meets or exceeds the value required by $V_{IC(max)}$:

$$S_{3(req)} = \frac{2I_{D3}}{K_P'(V_{DD} - V_{IC(max)} + V_{TN0})^2} = \frac{2 \times 125\mu}{63.8\mu \times (1.2 - 1.05 + 0.343)^2} = \frac{250}{63.8 \times 0.2430} \approx 16.12 \tag{18}$$

Since $S_3 = 174.14 \gg 16.12$, the condition is **satisfied**. With $S_3 = 174.14$, the actual $V_{IC(max)}$ can exceed 1.05 V by a wide margin.

2.8 Step 8: Differential Voltage Gain ($A_v = g_{m0} R_{out}$)

Following **Example 24-4 (page 24-22)**, g_m and g_{ds} of each transistor group are computed at its **quiescent current**:

$$M_3, M_4\ (I = 125\mu\text{A}): \quad g_{m4} = \sqrt{2 \times 125\mu \times 63.8\mu \times 174.14} \approx 1665\ \mu\text{S}$$
$$g_{ds4} = 125\mu \times 0.0469 = 5.86\ \mu\text{S} \tag{19}$$

$$M_5, M_6\ (I = 75\mu\text{A}): \quad g_{m6} = \sqrt{2 \times 75\mu \times 63.8\mu \times 174.14} \approx 1290\ \mu\text{S}$$
$$g_{ds6} = 75\mu \times 0.0469 = 3.52\ \mu\text{S} \tag{20}$$

$$M_7 \ldots M_{10}\ (I = 75\mu\text{A}): \quad g_{m8} = \sqrt{2 \times 75\mu \times 457.2\mu \times 24.30} \approx 1290\ \mu\text{S}$$
$$g_{ds8} = g_{ds10} = 75\mu \times 0.0528 = 3.96\ \mu\text{S} \tag{21}$$

$$M_0, M_1\ (I = 50\mu\text{A}): \quad g_{mI} = \sqrt{2 \times 50\mu \times 457.2\mu \times 8.63} \approx 628.3\ \mu\text{S}$$
$$g_{ds1} = 50\mu \times 0.0528 = 2.64\ \mu\text{S} \tag{22}$$

Resistance looking into the lower branch (R_{II}, Allen's $g_{m9}r_{ds9}r_{ds11}$):

$$R_{II} \approx g_{m8}r_{ds8}r_{ds10} = (1290\ \mu\text{S})\left(\frac{1}{3.96\ \mu\text{S}}\right)\left(\frac{1}{3.96\ \mu\text{S}}\right) \approx 82.3\ \text{M}\Omega \tag{23}$$

Output resistance (R_{out}):

$$\begin{aligned} R_{out} &\approx R_{II} \parallel (g_{m6}r_{ds6})\left(\frac{1}{g_{ds4} + g_{ds1}}\right) \\ &= 82.3\text{M} \parallel \left(\frac{1290}{3.52}\right)\left(\frac{1}{5.86 + 2.64}\right)\text{M}\Omega \\ &= 82.3\text{M} \parallel (366.5)\left(\frac{1}{8.50\ \mu\text{S}}\right) = 82.3\text{M} \parallel 43.1\text{M} \approx 28.3\ \text{M}\Omega \end{aligned} \tag{24}$$

Small-signal differential voltage gain:

$$\boxed{A_{vd} = g_{mI}R_{out} = (628.3\ \mu\text{S})(28.3\ \text{M}\Omega) = \mathbf{17781}\ \text{V/V}\ (\approx \mathbf{85.0}\ \text{dB}) \geq 74\ \text{dB}} \quad (\checkmark) \tag{25}$$

The hand estimate is well above the required 5000 V/V, as expected from the cascode output stage.

2.9 Step 9: Power Dissipation

The total current drawn from V_{DD} is $I_{D3} + I_{D4} = 125\mu + 125\mu = 250\ \mu\text{A}$:

$$\boxed{P_{diss} = (V_{DD} - V_{SS})(I_{D3} + I_{D4}) = 1.2 \times 250\mu = \mathbf{300}\ \mu\text{W}} \tag{26}$$

3 Bias Voltages

The schematic needs four DC bias voltages: V_{N1}, V_{P1}, V_{P2} and V_{N2}. They are set from the overdrive voltages (V_{OV}) of the corresponding transistors so that every device stays in saturation.

Overdrive voltages at the quiescent point:

$$V_{OV2} = \sqrt{\frac{2I_{D2}}{K'_N S_2}} = \sqrt{\frac{2 \times 100\mu}{457.2\mu \times 3.61}} \approx 0.348\ \text{V} \tag{27}$$

$$|V_{OV3}| = \sqrt{\frac{2I_{D3}}{K'_P S_3}} = \sqrt{\frac{2 \times 125\mu}{63.8\mu \times 174.14}} \approx 0.150\ \text{V} \tag{28}$$

$$|V_{OV5}| = \sqrt{\frac{2I_{D5}}{K'_P S_5}} = \sqrt{\frac{2 \times 75\mu}{63.8\mu \times 174.14}} \approx 0.116\ \text{V} \tag{29}$$

$$V_{OV7} = V_{OV9} = \sqrt{\frac{2I_{D7}}{K'_N S_7}} = \sqrt{\frac{2 \times 75\mu}{457.2\mu \times 24.30}} \approx 0.116\ \text{V} \tag{30}$$

3.1 V_{N1} (gate of M_2)

Sets the tail current $I_{D2} = 100\ \mu\text{A}$:

$$V_{N1} = V_{SS} + V_{TN0} + V_{OV2} = 0 + 0.343 + 0.348 = \mathbf{0.691}\ \text{V} \tag{31}$$

3.2 V_{P1} (gates of M_3, M_4)

Sets the branch current $I_{D3} = 125\ \mu$A:

$$V_{P1} = V_{DD} - (|V_{TP0}| + |V_{OV3}|) = 1.2 - (0.235 + 0.150) = \mathbf{0.815}\ \text{V} \tag{32}$$

3.3 V_{P2} (gates of M_5, M_6)

To maximize the output swing (wide-swing cascode), V_{SD3} is set to $|V_{OV3}| = 0.150$ V. The source of M_5 is then at $V_{DD} - 0.150 = 1.050$ V.

$$V_{P2} = V_{DD} - |V_{OV3}| - (|V_{TP0}| + |V_{OV5}|) = 1.2 - 0.150 - (0.235 + 0.116) = \mathbf{0.699}\ \text{V} \tag{33}$$

3.4 V_{N2} (gates of M_7, M_8)

Likewise, for the lower swing V_{DS9} is set to $V_{OV9} = 0.116$ V, so the source of M_7 is at 0.116 V.

$$V_{N2} = V_{SS} + V_{OV9} + V_{TN0} + V_{OV7} = 0 + 0.116 + 0.343 + 0.116 = \mathbf{0.575}\ \text{V} \tag{34}$$

Check of the self-biased bottom mirror: in the schematic, the gates of M_9, M_{10} are tied to the drain of M_7 (node V_{D7}). The circuit settles at $V_{D7} = V_{GS9} = V_{TN0} + V_{OV9} = 0.459$ V. Checking that M_7 is in saturation: $V_{DS7} = V_{D7} - V_{S7} = 0.459 - 0.116 = 0.343$ V. Since $V_{DS7} = 0.343\ \text{V} > V_{OV7} = 0.116$ V, M_7 is well inside saturation, so the current mirrored through M_9, M_{10} is accurate and the mirror has a high output resistance.

4 Summary

Transistor	$S = W/L$	W **(μm)**	I_D **(μA)**	**Set by**
M_0, M_1 (input pair)	8.63	4.32	50	Step 5 (GB)
M_2 (tail)	3.61	1.81	100	Step 6 (ICMR min)
M_3, M_4 (top PMOS)	174.14	87.1	125	Step 3 (V_{out} max)
M_5, M_6 (PMOS cascode)	174.14	87.1	75	Step 3 (worst case)
M_7, M_8 (NMOS cascode)	24.30	12.15	75	Step 4 (worst case)
M_9, M_{10} (NMOS mirror)	24.30	12.15	75	Step 4 (V_{out} min)